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MSc in Information and Communications Engineering,
Autonomous Systems and Robotics
The University of Klagenfurt, Klagenfurt/ Austria



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🌐 Portfolio

PROFESSIONAL SUMMARY

Autonomous Systems & AI Engineer with production deployment experience at Infineon Technologies and research validated in real field missions. Delivered the team's first LLM-powered data parsing pipeline for industrial mixed-signal verification; led end-to-end UAV system development and deployment, and developed AI-based landing site detection for the Austrian Space Forum's AMADEE-24 Mars analogue mission, validated offline on mission area data. MSc in Autonomous Systems & Robotics (University of Klagenfurt, graduated July 2026), OeAD Scholarship holder.

EXPERIENCE

•Infineon Technologies

Feb 2025-Jan 2026

Internship: Automated Verification Flow Development

Villach/ Austria

- Verification Workflow Enhancement & Standardization: Led the optimization of the mixed-signal verification flow by refactoring legacy code to be more robust, generalized, and user-friendly. Engineered key new features including automated report generation and direct regression execution capabilities.
- LLM-Integrated Data Parsing: Implemented Large Language Models to modernize data parsing workflows, establishing validation protocols to ensure data trustworthiness and integrity before downstream processing.

•University of Klagenfurt, Control of Networked Systems (CNS) Group

May 2023-December 2023

Research Student

Klagenfurt/ Austria

- Participated in AMADEE-24 Analogue Mars Mission led by Austrian Space Forum (ÖWF) as a team leader of RAMSES.
- Developed AI-based autonomous landing site detection in an unknown environment.
- Built simulation environments with various scenarios for implementing computer vision tasks in aerial vehicles by data gathering from environments.

•Adnan Menderes Airport, DHMİ (General Directorate of State Airports Authority)

June 2019-July 2019

Volunteer Internship

Izmir/Turkey

- Worked at the Electronics Department of DHMİ for maintenance of navigation devices.
- Have been involved in researching the network connection project of the navigation devices with the headquarters.

•ABK Cesme RES Inc.

Jan 2019-Feb 2019

Compulsory Internship

Izmir/Turkey

- Analyzed stages of wind turbines investment project management.
- Assisted with power line selection and simulation of wind turbines to how much energy they produce.
- Worked on the electrical project of designing the area of the power station center.

EDUCATION

•University of Klagenfurt, Klagenfurt/Austria

2022-July 2026

MSc Information and Communications Engineering branch of Autonomous Systems and Robotics

GPA: 2.04/5.0 (1.0 = best)

- OeAD and University of Klagenfurt Technology Scholarship holder.

•Izmir University of Economics, Izmir/Turkey

2015-2020

BSc Electric and Electronics Engineering

CGPA/Percentage: 3.08/4.00

- Graduated with Honor Roll.

•Rahmi Kula Anatolian High School, Balıkesir/Turkey

2011-2015

Science and Mathematics branch

CGPA/Percentage: 85.19/100

PROJECTS

•Master Thesis in the Control of Networked Systems group at the University of Klagenfurt

August 2023-July 2026

Master thesis on "Precise Landing Site Detection using Deep Learning for UAVs" involves developing a system that allows Unmanned Aerial Vehicles (UAVs) to accurately identify suitable landing sites using advanced deep learning techniques.

- Developing a deep learning network that includes semantic segmentation and optical flow models by joint learning.
- Pipeline for precise detection of landing sites in the environments using the network's output.
- Testing the network and pipeline on data generated from Isaac Sim.
- Testing the network and pipeline on data gathered from AMADEE-24 Analogue Mars Mission.
-  Thesis  Repo

•AMADEE-24 Analogue Mars Mission led by Austrian Space Forum (ÖWF)

August 2022-November 2024

Participating in AMADEE-24 Analogue Mars Mission as a team leader of the Rover Aerial Mars Support and Exploration System (RAMSES). The RAMSES is a technology system for future UAVs using an AI-based network to autonomously detect landing sites in Mars-like environments.

- Assembled aerial vehicle (hardware and software) and deploying CNS Flight Stack into the aerial vehicle.
- Developed user interface for receiving user-defined trajectory.
- Offline autonomous landing site detection evaluated on Mars-like environment data gathered with a micro aerial vehicle.
- Provided a better perception of scientific sites of interest leveraging the landing detector to investigate Mars-like environments.
- Worked on the aerial image data and orthoimages data to provide the coordinates of the notable scientific features supplied by the network.
- Featured in  AMADEE-24 Experiment Spotlight: RAMSES Autonomous Helicopter System, an interview covering the RAMSES UAV system.

•Drone Mars Simulation

October 2023-June 2024

Aerial vehicles are developed in simulation environments for accurate synthetic data. This offers cost-saving benefits, safety improvements, and closely mimics real-world scenarios testing. Creating accurate simulation environments is crucial for training deep learning models and improving autonomous systems.

- Used NVIDIA Omniverse Isaac Sim to create Mars-like environments.
- Gathered segmentation, depth, and IMU data for using scientific computer vision tasks.
- Adapted aerial vehicles in a simulation environment to test various robotic tasks.

•Graduation Project "Agricultural Autonomous Robot" Izmir University of Economics, Department of Electrical and Electronics Engineering.

Sept 2019-June 2020

The development of the Turtlebot3 Burger, a mobile educational robot, has led to the creation of a multi-sensor device capable of autonomously navigating through orchard environments.

- Working with the camera as a sensor for calculating the distance from the robot to object: Python and LabVIEW.
- Using Kalman Filter with HC-SR04 ultrasonic sensor's data for State Estimation.
- Collaboration with an industrial company.
- Implemented SLAM-based autonomous navigation on TurtleBot3 using ROS, enabling real-time localization and mapping in simulation environments.

TECHNICAL SKILLS AND INTERESTS

Languages: English: Advanced. German: Pre-Intermediate. Italian: Beginner. Turkish: Native.

Robotics & Autonomy: ROS1/2, PX4, SLAM, VIO, visual odometry, sensor fusion (IMU, depth cameras), autonomous navigation, state estimation (EKF, UKF, IEKF), UAV assembly & deployment, Robot Data Pipelines (PostgreSQL, MobilityDB, Grafana).

IT Skills: MS Office Programs, MATLAB, Proteus, MicroC PRO for PIC and C, C++, Python, LabView, CST Studio Suite, Git, Docker, AWS, IsaacSim.

DL/ML: Experienced in Deep Learning and Machine Learning (semantic segmentation, object detection, GANs, optical flow). Proficient in PyTorch and TensorFlow, and skilled in building web applications with Flask, Gradio, and FastAPI. Experience in developing CI/CD pipelines within AWS and GitHub Actions.

Large Language Models (LLMs) & Vision-Language Models (VLMs): Applied LLMs for automated data parsing pipelines with validation protocols (RAG systems, local inference agents). Integrated VLMs for semantic scene understanding in robotic navigation, linking visual perception to robot actions via language interfaces.

Areas of Interest: Computer vision applications in robotics, UAV/Mobile robots. Image processing. Visual robotics navigation. Deep learning and machine learning applications e.g. computer vision and state estimation. Data analysis for scalable AI pipelines. Deep Learning & LLM integration for industrial applications.

ACHIEVEMENTS & ACTIVITIES

- Presentation:** Presented experimental outcomes of the RAMSES project at the AMADEE-24 Science Workshop (ÖWF) in Salzburg. Nov 2024
- Volunteering:** Referee at RoboCupJunior Austrian Open. Oversaw competition rules and scored team performances. April 2023
- HACK'O LIFE Hackathon:** Served on the organizing committee of HACK'O LIFE, a hackathon event held at Izmir University of Economics. Coordinated event logistics, mentored participating teams throughout the competition, and provided technical guidance to help teams develop and present their projects. 2019
- UAM Basketball 3x3 — Unisport Austria Meisterschaft:** Represented the University of Klagenfurt at the national university basketball championship (3x3 format) organised by Unisport Austria. Competed as part of the university team in the men's category. 2025, 2026
- Events:** Participant in Izmir Energy Summit (2019); Work & Travel USA (2017). 2017-2019

INDIVIDUAL PROJECTS

- AI/CV Product Development:** Building a multimodal AI/computer vision proof-of-concept for textile anomaly detection as part of an early-stage venture, currently pursuing funding. Leading technical architecture and development. AI Product Ongoing
- Multi-Sensor Fusion for GNSS-Denied UAV Navigation:** Designed and implemented a custom MultiModal EKF fusing IMU (200 Hz), Barometer, Magnetometer, Visual Odometry (RTAB-Map), and LiDAR ICP for GPS-denied autonomous flight. Built on PX4 SITL + Gazebo Harmonic + ROS2 Humble, featuring runtime sensor toggling, trajectory logging, and a PyQt5 fusion monitor GUI with integrated mission control.  **Repo** UAV Nav. Ongoing
- Edge AI for Agricultural Robotics:** Optimizing a multi-model leaf disease detection pipeline (YOLO, Mask R-CNN) for real-time deployment on UAVs and mobile robots. Converting models to ONNX format and applying quantization techniques to ensure low-latency inference on resource-constrained embedded edge devices. Edge AI & Robotics Ongoing
- VLM-Based Semantic Navigation:** Engineered a custom SemanticCostmapLayer plugin for Nav2, converting VLM (LLaVA/Moondream) scene descriptions into local semantic hazards. Orchestrated perception workflows via Ollama and Gazebo Harmonic, transitioning from a legacy stop/go supervisor to dynamic costmap augmentation.  **Repo** Autonomous Nav. Ongoing

- Leaf Disease Detection:** Developed an automated diagnostic pipeline to identify diseases and produce types by fusing outputs from DeepLabV3, Mask R-CNN, and YOLO. Integrated SAM3 for interactive, text-prompted segmentation and deployed the solution using Flask and Docker.  **Repo** *Computer Vision
Completed*
- Agricultural Disease Dataset Aggregation:** Collecting and unifying publicly available leaf disease datasets across segmentation, classification, and object detection tasks to expand class diversity and training data volume. Future plan: structure into COCO format and manage metadata via a cloud-hosted PostgreSQL database. *AgriAI
Ongoing*
- Home Indoor Hydroponic Monitoring Dashboard:** Developed a real-time basic IoT dashboard for an indoor hydroponic system using Flask, Flask-SocketIO, and Chart.js. Deployed on a Raspberry Pi to read DHT22 and MCP3008 sensors via GPIO/SPI, featuring automated threshold alerts and REST APIs. *IoT & Embedded
Ongoing*

MSc/BSc COURSE PROJECTS

- MSc Project:** AI for Mobile Robots: Engineered a bidirectional LSTM (PyTorch) to predict 2D trajectories from 6D IMU data and trained a CNN-based Deep Q-Network (DQN) for reinforcement learning tasks. *Mob. Robot Nav.
22W*
- MSc Project:** Vision-Based State Estimation: Developed a full Visual Odometry (VO) pipeline including camera calibration, 3D triangulation, feature extraction (FAST, Harris), and RANSAC localization. Implemented UKF and IEKF for nonlinear state estimation (Python, MATLAB). *VBSE
22W*
- MSc Project:** Artificial Vision: Developed a leaf disease detection pipeline utilizing classical computer vision methods and applied Grad-CAM to generate interpretable heatmaps for classification decisions. *Artificial Vision
22S*
- MSc Project:** ML in Communications: Developed Neural Decoders for Hamming(7,4) codes, built end-to-end communication systems using AutoEncoders, and modeled AWGN channels using CGANs (Python, TensorFlow). *ML for ICE
22S*
- MSc Project:** Autonomous Driving Lab: Implemented a full perception and control pipeline in CARLA simulator, including lane detection, target point estimation, traffic sign classification (CNN), and semantic segmentation. *ADC Lab
23S*
- MSc Project:** Generative AI: Optimized DCGAN and Conditional DCGAN models on CIFAR-10 dataset (Python, TensorFlow). One-Shot Learning: Analyzed and implemented Siamese Neural Networks on MNIST dataset (Python, TensorFlow). *ML in Intel. Trans.
22S*

REFERENCES

- Assoc. Prof. Dr. Jan Steinbrener (University of Klagenfurt)**
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Formal reference letter available upon request.
- Dr. Thomas Georg Jantos, M.Sc. (University of Klagenfurt)**
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Formal reference letter available upon request.
- Infineon Technologies Austria AG, Villach**
Design Enabling & Services, Analog Centric IP Verification
Formal reference letter available upon request.